**Learning Objective 9.5:** I can determine the reflection and transmission of UPW at a boundary for Transverse Electric (TE) and Transverse Magnetic (TM) waves, using boundary conditions.

Show your work. Write a **Documentation** line at the end (**None** if you did not collaborate). In every problem the boundary is the plane $z = 0$, medium 1 is $z < 0$, medium 2 is $z > 0$, the incident wave travels in $+z$ with $\widetilde{\mathbf{E}}_i = \hat{\mathbf{x}} E_0 e^{-j\beta_1 z}$, and every medium is nonmagnetic, $\mu = \mu_0$.

1. **Deriving the coefficients.** Two lossless media meet at $z = 0$, with intrinsic impedances $\eta_1$ and $\eta_2$.

   (a) Write the electric and magnetic fields of the incident, reflected, and transmitted waves in terms of $E_0$, $\Gamma$, and $\tau$.

   (b) Apply the continuity of tangential **E** and tangential **H** at $z = 0$, and solve for $\Gamma$ and $\tau$.

   (c) Show that the reflected and transmitted fractions of the incident power, $|\Gamma|^2$ and $(\eta_1/\eta_2)\,|\tau|^2$, add to one.

2. **Air to glass.** A 3 GHz plane wave with $E_0 = 10$ V/m travels in air and strikes glass with $\epsilon_r = 4$.

   (a) Find $\eta_2$, $\Gamma$, and $\tau$.

   $\Gamma$ =

   $\tau$ =

   (b) Find the amplitudes of the incident, reflected, and transmitted magnetic fields, and show that tangential **H** is continuous.

   $H_t$ =

   (c) Find the SWR in the air, the largest and smallest field magnitudes, and the distance from the boundary to the nearest maximum.

   SWR =

   max at =

   (d) Find the incident power density and the percentages of the power that reflect and transmit.

   reflected =

3. **Glass to air.** The same boundary, now with the wave arriving from inside the glass: medium 1 has $\epsilon_r = 4$, and medium 2 is air.

   (a) Find $\Gamma$ and $\tau$.

   $\tau =$ ____

   (b) The transmitted electric field is larger than the incident one. Explain why that does not create power, and find the fraction of the power transmitted.

   (c) Where is the field in the glass largest, and why does the answer differ from Question 2?

4. **A conducting wall.** A 10 GHz plane wave with $E_0 = 5$ V/m (real) travels in air and strikes a flat perfect conductor that fills $z > 0$.

   (a) Use the boundary condition on tangential **E** to find $\Gamma$, and write the total phasor fields in the air.

   (b) Find the distance from the wall to the nearest maximum of $|\mathbf{E}|$, and its value.

   distance = ____

   (c) Find the surface current density on the wall.

   $\mathbf{J}_s$ = ____

   (d) Write $\mathbf{E}(z,t)$ and $\mathbf{H}(z,t)$, and explain why no power flows toward the wall on average.

**Documentation:** _____